Appendix A. List of papers addressing recommendations on AI tools applied in introductory programming teaching.

R1. Karvelas I, Li A, Becker BA (2020) The effects of compilation mechanisms and error message presentation on novice programmer behavior. SIGCSE 2020 - Proceedings of the 51st ACM Technical Symposium on Computer Science Education 759–765. https://doi.org/10.1145/3328778.3366882

R2. Chen S-J, Shan X, Liu Z-M, Chen C-Q (2025) Employing large language models to enhance K-12 students' programming debugging skills, computational thinking, and self-efficacy. Technology & Society 28:259–278. https://doi.org/10.31124/ADVANCE.171198179.96624107

R3. Finnie-Ansley J, Denny P, Becker BA, et al (2022) The robots are coming: Exploring the implications of OpenAI codex on introductory programming. ACM International Conference Proceeding Series 10–19. https://doi.org/10.1145/3511861.3511863

R4. Kazemitabaar M, Hou X, Henley AZ, et al (2023) How Novices Use LLM-Based Code Generators to Solve CS1 Coding Tasks in a Self-Paced Learning Environment. ACM International Conference Proceeding Series 1–12. https://doi.org/10.1145/3631802.3631806

R5. Prather J, Denny P, Leinonen J, et al (2023) The Robots are Here: Navigating the Generative AI Revolution in Computing Education. ITiCSE-WGR 2023 - Proceedings of the 2023 Working Group Reports on Innovation and Technology in Computer Science Education 108–159. https://doi.org/10.1145/3623762.3633499

R6. Denny P, Leinonen J, Prather J, et al (2024) Prompt Problems: A New Programming Exercise for the Generative AI Era. SIGCSE 2024 - Proceedings of the 55th ACM Technical Symposium on Computer Science Education 1:296–302. https://doi.org/10.1145/3626252.3630909

R7. Styve A, Virkki OT, Naeem U (2024) Developing Critical Thinking Practices Interwoven with Generative AI Usage in an Introductory Programming Course. IEEE Global Engineering Education Conference, EDUCON 01–08. https://doi.org/10.1109/EDUCON60312.2024.10578746

R8. Ma Q, Shen H, Koedinger K, Wu ST (2024) How to Teach Programming in the AI Era? Using LLMs as a Teachable Agent for Debugging. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 14829 LNAI:265–279. https://doi.org/10.1007/978-3-031-64302-6_19

R9. Sheese B, Liffiton M, Savelka J, Denny P (2024) Patterns of Student Help-Seeking When Using a Large Language Model-Powered Programming Assistant. ACM International Conference Proceeding Series 49–57. https://doi.org/10.1145/3636243.3636249

R10. Fernandez AS, Cornell KA (2024) CS1 with a Side of AI: Teaching Software Verification for Secure Code in the Era of Generative AI. SIGCSE 2024 - Proceedings of the 55th ACM Technical Symposium on Computer Science Education 1:345–351. https://doi.org/10.1145/3626252.3630817

R11. Prather J, Denny P, Leinonen J, et al (2024) Interactions with Prompt Problems: A New Way to Teach Programming with Large Language Models. ArXiv 1–30

R12. Hoq M, Shi Y, Leinonen J, et al (2024) Detecting ChatGPT-Generated Code Submissions in a CS1 Course Using Machine Learning Models. SIGCSE 2024 - Proceedings of the 55th ACM Technical Symposium on Computer Science Education 1:526–532. https://doi.org/10.1145/3626252.3630826

R13. Prather J, Reeves BN, Leinonen J, et al (2024) The Widening Gap: The Benefits and Harms of Generative AI for Novice Programmers. ICER 2024 - ACM Conference on International Computing Education Research 1:469–486. https://doi.org/10.1145/3632620.3671116

R14. Zviel-Girshin R (2024) The Good and Bad of AI Tools in Novice Programming Education. Educ Sci (Basel) 14:. https://doi.org/10.3390/educsci14101089

R15. Huang AYQ, Lin CY, Su SY, Yang SJH (2025) The impact of GenAI-enabled coding hints on students' programming performance and cognitive load in an SRL-based Python course. British Journal of Educational Technology 56:1942–1972. https://doi.org/10.1111/bjet.13589